```
> restart:

  VerifyLemma22 := proc()
      local Tmin, Tmax, Nmax, Ptab, Atab,
            n, r, g, sgn, total, t, det,
            minDa, minAt, checks;

      Tmin := 72;
      Tmax := 9999;
      Nmax := Tmax+1;

      # Use exact Maple integers.
      Ptab := Array(0..Nmax, datatype=anything, fill=0);
      Ptab[0] := 1;

      # Euler's pentagonal recurrence for p(n).
      for n from 1 to Nmax do
          total := 0;
          r := 1;
          sgn := 1;

          while true do
              g := iquo(r*(3*r-1), 2);

              if g > n then
                  break;
              end if;

              total := total+sgn*Ptab[n-g];

              if g+r <= n then
                  total := total+sgn*Ptab[n-g-r];
              end if;

              r := r+1;
              sgn := -sgn;
          end do;

          Ptab[n] := total;
      end do;

      # Compute a(t) = p(t)-p(t-1).
      Atab := Array(1..Nmax, datatype=anything, fill=0);

      for t from 1 to Nmax do
          Atab[t] := Ptab[t]-Ptab[t-1];
```

```
    end do;

    # Check every determinant and retain its first minimum.
    minDa := 0;
    minAt := 0;
    checks := 0;

    for t from Tmin to Tmax do
        det := Atab[t]^2-Atab[t-1]*Atab[t+1];

        if det <= 0 then
            error "Nonpositive determinant: [t,D_a(t)]=%1",
                  [t,det];
        end if;

        if t = Tmin or det < minDa then
            minDa := det;
            minAt := t;
        end if;

        checks := checks+1;
    end do;

    # Verify the exact certificate recorded in Appendix A.
    if checks <> 9928
       or minDa <> 28877078
       or minAt <> 73 then
        error "Certificate mismatch: [checks,minimum,index]=%1",
              [checks,minDa,minAt];
    end if;

    print(["LEMMA_2_2_FINITE_PART_PASS", Tmin, Tmax]);
    print(["NUMBER_OF_CHECKS", checks]);
    print(["MINIMUM_Da", minDa]);
    print(["FIRST_ATTAINED_AT", minAt]);

    return [checks,minDa,minAt];
end proc:

CertificateResult := VerifyLemma22():
```

```
["LEMMA_2_2_FINITE_PART_PASS", 72, 9999]
["NUMBER_OF_CHECKS", 9928]
["MINIMUM_Da", 28877078]
["FIRST_ATTAINED_AT", 73]
```

```
>
```